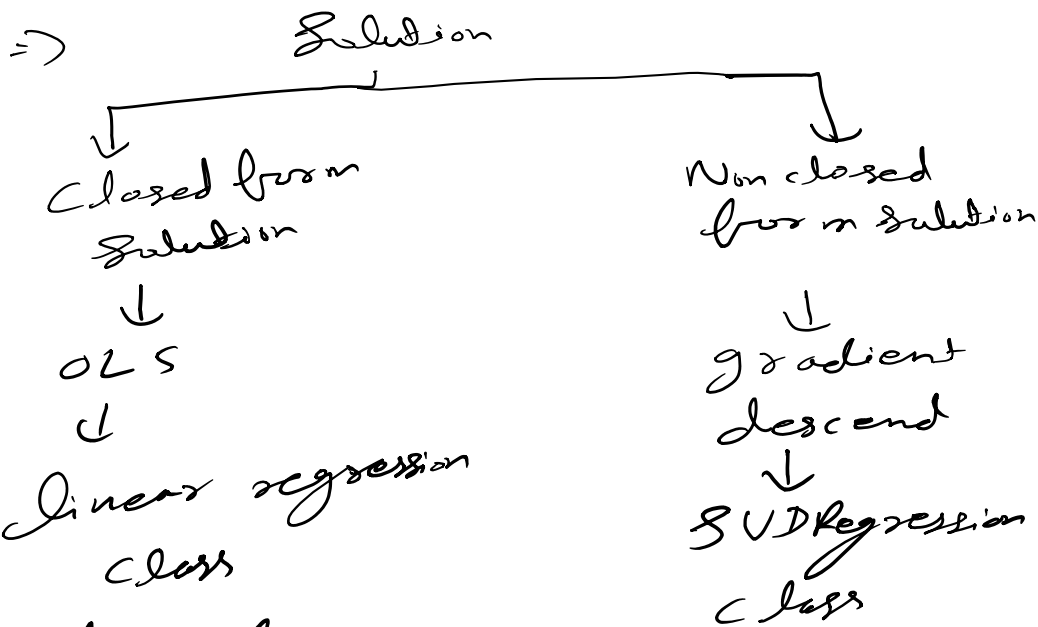


Linear Regression

Sunday, June 28, 2026 4:14 PM



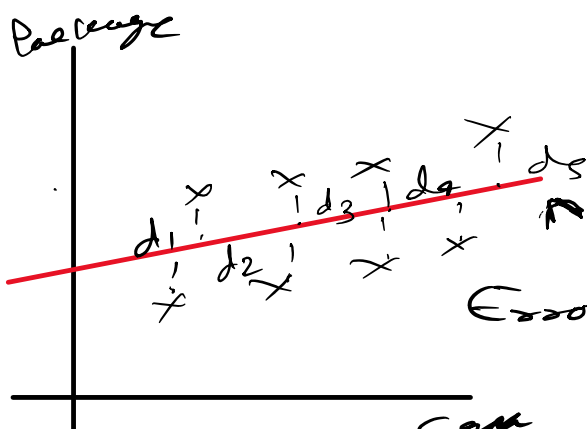
=> formula:-

$$b = \bar{y} - m\bar{x}$$

$\bar{x}$ → Cgpa → Package
 $\bar{y}$ → mean

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Cgpa	Package
6.5	0
7.2	1
8.4	0
5.2	1



$$\text{Error} = d_1 + d_2 + d_3 + d_4$$

↓ Squaring

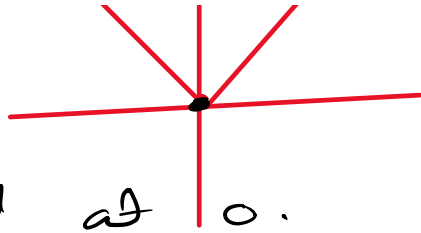
So it is positive

$$\Rightarrow E = d_1^2 + d_2^2 + d_3^2 + d_4^2$$

=> why do not take mode:-

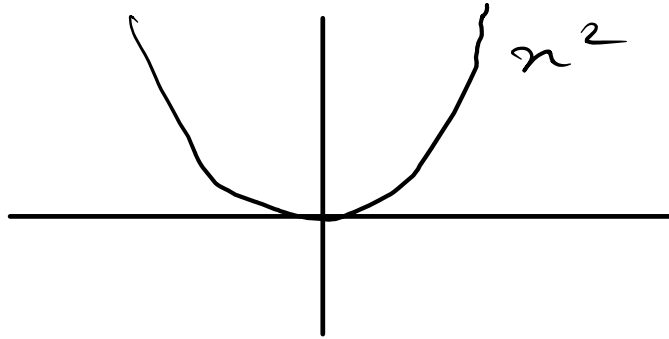
Because

mode is not



defined at 0.

⇒ we want continuous data:-



⇒ Hence Error:-

$$E = \sum_{i=1}^n d_i^2$$

$$d_i = y_i - \hat{y}_i$$

$$E = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

(m, b) ↗

$$\hat{y}_i = m x_i + b$$

$$E = \frac{1}{n} \sum_{i=1}^n (y_i - m x_i - b)^2$$

↪ m, b choose that way so
 ... the error is

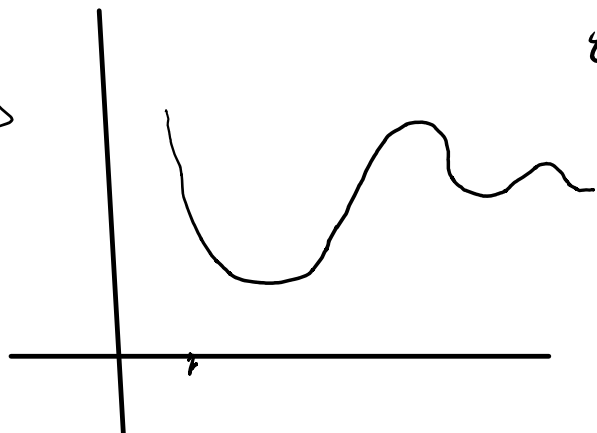
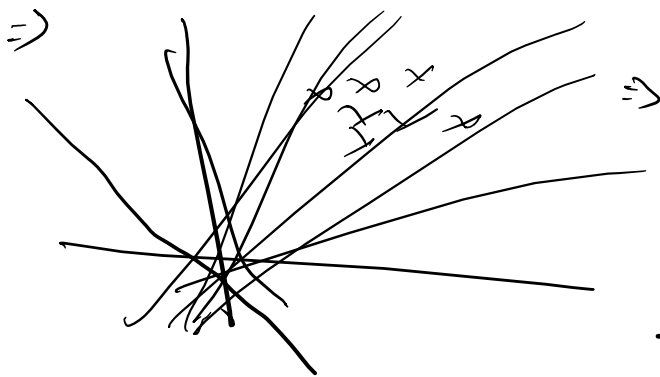
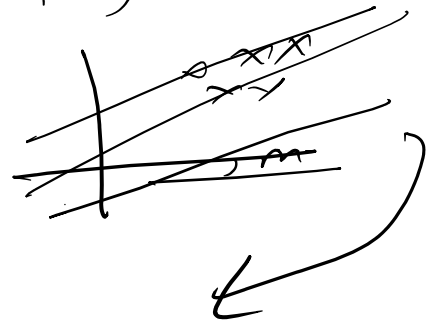
$\hookrightarrow m, b$ chosen to
 decrease the error or
 minimize the distance.

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

(m, b)
 $\downarrow$
 $b = 0$

$$E(b) = \sum_{i=1}^n (y_i - x_i - b)^2$$

$$E(m) = \sum_{i=1}^n (y_i - mx_i)^2$$



$\Rightarrow$ Maxima and Minima:—

$$\begin{aligned}
 \frac{\partial E}{\partial b} &= \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - mx_i - b)^2 \\
 &= \sum \frac{\partial}{\partial b} (y_i - mx_i - b)^2 \\
 &= \sum 2 \times (y_i - mx_i - b) \cdot (-1) \\
 &= -2 \sum (y_i - mx_i - b) = 0
 \end{aligned}$$

$$= \sum (y_i - mx_i - b)$$

$$= \sum (y_i - mx_i - b) = 0$$

$$= \sum y_i - \sum mx_i - \sum b = 0$$

$\Rightarrow$ divide by n

$$\Rightarrow \frac{\sum y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum b}{n} = 0$$

$$\Rightarrow \bar{y} - m\bar{x} - \frac{nb}{n} = 0 \quad \left| \begin{array}{l} \sum b = b+b+b+ \\ b+\dots+b \\ \hline n \end{array} \right.$$

$$\Rightarrow \bar{y} - m\bar{x} = b$$

$$\Rightarrow E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$= 2 (y_i - mx_i - \bar{y} + m\bar{x})$$

$$= (-x_i + \bar{x})$$

$$\Rightarrow \sum (y_i - mx_i - \bar{y} + m\bar{x})$$

$$\sum (-x_i + \bar{x}) = 0$$

$$\Rightarrow \sum [(y_i - \bar{y}) - m(x_i - \bar{x})] (x_i - \bar{x}) = 0$$

$$\Rightarrow \sum [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2]$$

$$\Rightarrow \sum [(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2] = 0$$

$$\Rightarrow \sum (y_i - \bar{y})(x_i - \bar{x}) = \sum m(x_i - \bar{x})^2$$

$$\Rightarrow m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$